Actual S4: the first resultant step

For any nonsingular general Weierstrass curve over an algebraically closed field of any characteristic, Lean checks the first fixed-degree resultant step: its vanishing is equivalent to four actual affine points with the prescribed x-coordinates summing to zero. Repeated coordinates and partial sums at infinity are included. [pcc:unverified]

Write the two S3 polynomials as $aT^2 + bT + c$ and $dT^2 + eT + f$. The definition uses Mathlib's actual degree $2/2$ Sylvester determinant; Lean computes its matrix and proves the following identity. [pcc:unverified]

$$\mathrm{Res}_{2,2} = (af - cd)^2 - (ae - bd)(bf - ce).$$

With first-pair distinct x-coordinates, the checked S3 factorization gives roots $r = x(P + Q)$ and $s = x(P - Q)$. The resultant becomes $a^2 S_3(x_3, x_4, r) S_3(x_3, x_4, s)$. A common root glues actual points equal up to negation. The converse constructs the common coordinate from the nonzero actual pair sum. [pcc:unverified]

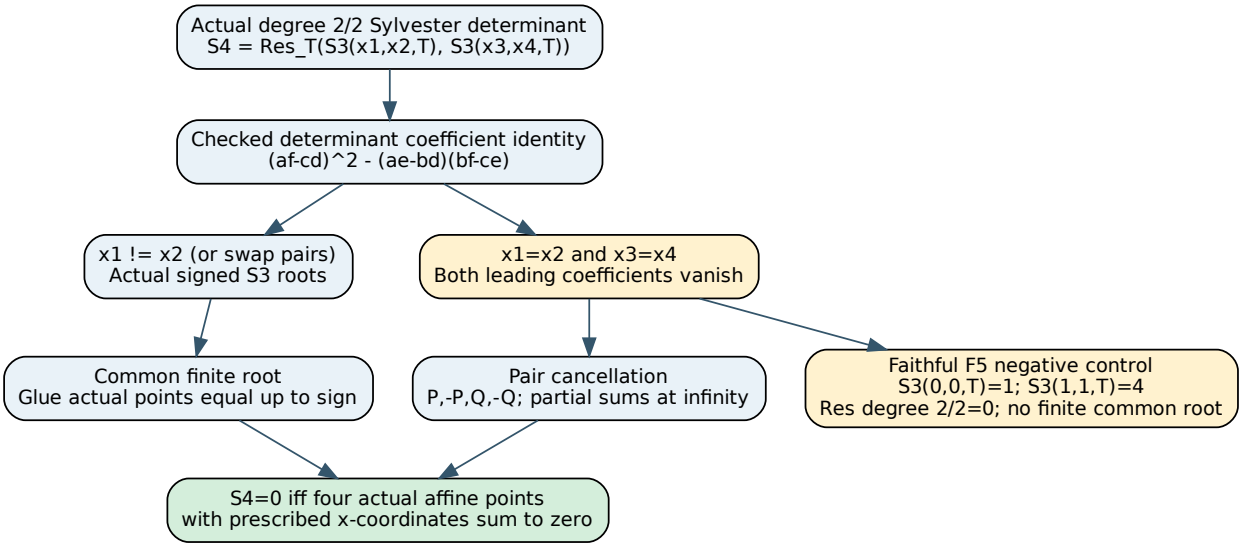


Figure 1: Proof dependencies; all scalar and point operations are Mathlib objects.

If the first pair repeats but the second does not, swap the pairs. If both repeat, both leading coefficients vanish and the fixed-size determinant is zero. Actual points $P, -P, Q, -Q$ provide the four-point sum; no finite intermediate coordinate is needed. This branch must not be replaced with an assumed finite-common-root criterion. [pcc:unverified]

Main source: ../../lean/GeneralS4.lean, declaration `GeneralS4Audit.generalS4_zero_iff_exists_points`. This establishes the $m = 4$ vanishing equivalence at every specialization; general recursion for all m, irreducibility and symbolic individual-variable degrees remain open. [pcc:unverified]

A geometric control for the degree-drop shortcut

Take the actual smooth curve $y^2 = x^3 - x$ over $\mathbf{F}_5$ and base change to its algebraic closure. Lean checks the following identities for every t in that closure. [pcc:unverified]

Quantity	Kernel statement
First specialized S3	$S_3(0,0,t) = 1$
Second specialized S3	$S_3(1,1,t) = 4$
Fixed degree $2/2$ resultant	$S_4(0,0,1,1) = 0$
Common finite root	Does not exist
Four actual affine points	A zero-sum quartet exists

The nonzero constants have no finite common root. The zero fixed-size resultant detects their common projective behavior at infinity. Recomputing a resultant using their new actual degrees would change the operation. This refutes the finite-root proof shortcut, while the qualified S4 point-sum equivalence is preserved. [pcc:unverified]

The determinant formula, factorization, group-law gluing, repeated-pair construction and negative control are kernel proofs, not finite-sampling inferences. The existing bounded S4/S5 census is unchanged. This increment runs no new curve population, benchmark or paid compute. [pcc:unverified]

Primary source read: Kosters–Yeo2015, Section 2/page 3 recursive definition and Proposition 2.1; PDF page visually checked. Exact digests/read scope: ../../general-s4-sources.json. Original Semaev 2004 remains unread: ePrint/CiteSeer returned 403 and the cited RHUL copy returned 404. No full-paper or all-fourteen completion is claimed. [pcc:unverified]

Whole census gate:197 declarations,0 failing,0 trusted-native exceptions. Packets PCC-20261109-21757f and PCC-20261109-ce6722 bind the theorem and control; full producer checks are retained separately and remain unsigned. The count includes generated declarations, not 197 new results. [pcc:unverified]

Supplement IDC1hbf1f44625ee306dc checks the determinant coefficient rewrite with the unchanged repository checker:32 derived plus11 extra points over $2^{61} - 1$,6 variables, degree bound 4. Source commit e6bcfc3a preceded issue; the proof-file digest matches GeneralS4.lean. One-host finite checks do not replace the universal Lean proof or establish separate-host standing. [pcc:unverified]